# Powershell Collection Troubleshooting

| | |
|---|---|
| Guide | credential gateway service, cyberark, WinRMTransportError, powershell_pywinrm_transport, authGSSClientStep, Unspecified GSS failure, EmptyPasswordError, kinit, KDC,Cannot contact any KDC for realm |
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- 2023-09-30 EM-61211: We are unable to discover windows servers
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- 2019-01-09 EM-34627: Windows server denying access for powershell credentials.

## Introduction

Here is a powerpoint where troubleshooting info has been gathered: [PowerShell Collections Troubleshooting.pptx]

And here is an article with more troubleshooting guidance: PowerShell Troubleshooting

## Diagnosis and Mitigation

### collection issues in ol8 collectors

See: https://sciencelogic.lightning.force.com/lightning/r/Knowledge__kav/ka0VL0000002yJRYAY/view

## theory

### config_file:/etc/krb5.conf

TBD

## Post Mortem Articles

### 2024-11-20 EM-71264: OL8 Powershell Collector issues

ts_guide_to_jira_link

Link: EM-71264: OL8 Powershell Collector issues CLOSED

From the description:

```
Observed Behavior: Powershell collection is failing for a number of servers due to TLS errors

Expected Behavior: Collection should work.

```

```
Summary of troubleshooting steps taken so far and the current status:

Customer has been seeing TLS errors after upgrading to OL8 collectors against devices.

They believe it is related to this KB article: https://sciencelogic.lightning.force.com/lightning/r/Knowledge__kav/ka0VL0000002yJRYAY/view
This was from the escalation
They are showing similar issues wit the powershell collector in FIPs mode:
They are having TLS errors when connecting as well:

error-2.png
credential.png

Would the dropfile from the other escalation help in this case?
Thank you
```

### 2024-10-30 EM-70933: PowerShell Events Randomly Triggers and Closing in 5 minutes

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Link: SOL-28902: PowerShell Events Randomly Triggers and Closing in 5 minutes CLOSED

From the description:

```
Observed Behavior:PowerShell Events Randomly Triggers and Closing in 5 minutes

Summary of troubleshooting steps taken so far and the current status:
Issue -
As per customer they are facing issue with Powershell Event Randomly triggered and close in 5 minutes.
We used to see [3618] PowerShell request error events for multiple devices and it getting closed within 5 minutes.
This event triggers for the Dynamic Application - Microsoft: SQL Buffer Performance (ID: 1616).
customer used to receive PowerShell events from dynamic applications: Microsoft: IIS Web Service Performance (ID: 215) and Microsoft: IIS Active Server Pages Performance ( ID: 217).

Observation -
The first described issue actually sounds like something that would be fixed by concurrent PS / pysprp, but was likely an enhancement to one of the collections applications due to the pack being updated to Python 3.
The Microsoft: IIS one, however, is a known issue we have seen before where the IIS apps would get automatically aligned to devices that did not run IIS. This would cause a flood of the messages. According to release notes, this was
fixed in version 102 of the Microsoft: IIS pack.
As per support instruction customer upgraded the Powerpacks of the below listed, but still the "PowerShell request error: App [1616]" is not resolved. We are getting around 4500+ notice events from this DA: 1616 - Microsoft: SQL Buffer Performance.
We enable the pypsrp and it took care of a other issues like cpu load, but now we have uncovered some problems with kerberos authentication.
Some initial sleuthing reveals this is affecting a handful of CUGs, but we stuck with the Germany one as an example.
The devices aligned to 01 in that group are all fine - however 02 and 03 are throwing the events reported below.
We have enabled PS debugging on 02 for now.
We observed below error on /etc/krb5.conf file I have asked customer to share this /etc/krb5.conf file  file.
```

```

<DOMAIN_NAME> = {
kdc  =  <Active_directory_hostname> # Failed to resolve this server by name/fqdn, verify DNS is correctly configured
kdc  =  <Active_directory_hostname> # Failed to resolve this server by name/fqdn, verify DNS is correctly configured
kdc  =  <Active_directory_hostname> # Failed to resolve this server by name/fqdn, verify DNS is correctly configured
kdc  =  <Active_directory_hostname> # Failed to resolve this server by name/fqdn, verify DNS is correctly configured
kdc  =  <Active_directory_hostname> # Failed to resolve this server by name/fqdn, verify DNS is correctly configured
```

### 2024-08-26 EM-66567: NWNA - PowerShell Communication Error on multiple Windows Server devices

ts_guide_to_jira_link

Link: EM-66567: NWNA - PowerShell Communication Error on multiple Windows Server devices CLOSED

From the description:

```
Observed Behavior: Powershell error: PowerShell Communication Error: App [2312] Windows device [172.17.3.76]: authGSSClientStep() failed:
(('Unspecified GSS failure. Minor code may provide more information', 851968),
("Cannot contact any KDC for realm 'GASCO.COM'", -1765328228))

Expected Behavior: Resolve the error

Summary of troubleshooting steps taken so far and the current status:
Customer reached out stating that they are getting PS communication error for multiple windows server belonging to NWNA-BND collector group.
Collector converted to OL8 and customer believes that the issue is related to below Jira:
JIRA: https://sciencelogic.atlassian.net/browse/EM-65572

Customer ran the PS test script and got below error:
I checked the credential test for the device:mqsynergiapp01.gasco.com and saw it wasn’t executing the 6th step:
It wasn’t executing below test:
Execute PowerShell Cmdlet. Attempts to execute the 'Get-WmiObject Win32_Process | Select Name' PowerShell Cmdlet using the credential.
Checking the device logs I can see below error:
PowerShell Communication Error: App [2301] Request [2867] Windows device [172.17.3.76]: LocalProtocol: Command did not respond within 90 seconds and was terminated (WinRMOperationTimeoutError: LocalProtocol: Command did not respond within 90 seconds and was terminated)
Suggested customer to increase the timeout from 90s to 180s and the errors can no longer be seen.
However, we can see below event in device logs:
PowerShell Communication Error: App [2312] Windows device [172.17.3.76]: authGSSClientStep() failed: (('Unspecified GSS failure. Minor code may provide more information', 851968), ("Cannot contact any KDC for realm 'GASCO.COM'", -1765328228))
I again performed credential test from UI and now it is passing:
I checked the configuration for kerberos and it looked correct:

[realms]
```

```
# http://EXAMPLE.COM  = {
# kdc = kerberos.example.com
# admin_server = kerberos.example.com
# }
LMS-MHG.LOCAL = {
kdc = lms-mhg.local
}
ADDS.NWNATURAL.COM = {
kdc = adds.nwnatural.com
}
GASCO.COM = {
kdc = gasco.com
}
GASCODMZ.COM = {
kdc = 192.168.210.250
kdc = 172.30.210.251
}
[domain_realm]
# .example.com = http://EXAMPLE.COM

# http://example.com  = http://EXAMPLE.COM
.gasco.com = GASCO.COM
gasco.com = GASCO.COM
.gascodmz.com = GASCODMZ.COM
gascodmz.com = GASCODMZ.COM
.adds.nwnatural.com = ADDS.NWNATURAL.COM
adds.nwnatural.com = ADDS.NWNATURAL.COM

I also compared this configuration with another device which had similar issue but now it has been resolved. The device for which issue has now been resolved is associated with another collector.
I further got below information winrm configuration:
Config
  MaxEnvelopeSizekb = 500
  MaxTimeoutms = 60000
  MaxBatchItems = 32000
  MaxProviderRequests = 4294967295
  Client
    NetworkDelayms = 5000
    URLPrefix = wsman
    AllowUnencrypted = false
    Auth
      Basic = true
      Digest = true
      Kerberos = true
      Negotiate = true
      Certificate = true
      CredSSP = false
    DefaultPorts
      HTTP = 5985
      HTTPS = 5986
    TrustedHosts
  Service
    RootSDDL = O:NSG:BAD:P(A;;GA;;;BA)(A;;GR;;;IU)S:P(AU;FA;GA;;;WD)(AU;SA;G
XGW;;;WD)
    MaxConcurrentOperations = 4294967295
    MaxConcurrentOperationsPerUser = 1500
    EnumerationTimeoutms = 240000
```

```
    MaxConnections = 300
    MaxPacketRetrievalTimeSeconds = 120
    AllowUnencrypted = false
    Auth
      Basic = false
      Kerberos = true
      Negotiate = true
      Certificate = false
      CredSSP = false
      CbtHardeningLevel = Relaxed
    DefaultPorts
      HTTP = 5985
      HTTPS = 5986
    IPv4Filter = * [Source="GPO"]
    IPv6Filter = * [Source="GPO"]
    EnableCompatibilityHttpListener = false
    EnableCompatibilityHttpsListener = false
    CertificateThumbprint = b3 56 4d b9 75 9d 80 67 7a 4b 7f 98 f6 31 91 5a

d9 0e 60 af
    AllowRemoteAccess = true [Source="GPO"]

  Winrs
    AllowRemoteShellAccess = true
    IdleTimeout = 7200000
    MaxConcurrentUsers = 2147483647
    MaxShellRunTime = 2147483647
    MaxProcessesPerShell = 2147483647
    MaxMemoryPerShellMB = 2147483647
    MaxShellsPerUser = 2147483647
PS C:\Windows\system32>


Compared the same with working device and wasn’t able to spot any difference.
After checking the collection for device mqsynergiapp01.gasco.com  I can see the collection has some missed polls, here is an example:
I further checked below information as mentioned in out TSG for the device:
\\Step 1
[root@NWNA-SHR-slc13 ~]# python -c "import socket;print(socket.gethostbyname('MQSynergiApp01.gasco.com'))"
172.17.3.76
[root@NWNA-SHR-slc13 ~]# python -c "import socket;print(socket.gethostbyaddr('172.17.3.76'))"
('MQSynergiApp01.gasco.com', [], ['172.17.3.76'])
\\Step 2
[root@NWNA-SHR-slc13 ~]# nmap -p 5985,5986 172.17.3.76
Starting Nmap 7.70 ( https://nmap.org ) at 2024-08-26 09:15 UTC
Nmap scan report for MQSynergiApp01.gasco.com (172.17.3.76)
Host is up (0.00021s latency).

PORT STATE SERVICE
5985/tcp open wsman
5986/tcp open wsmans
Nmap done: 1 IP address (1 host up) scanned in 0.49 seconds
[root@NWNA-SHR-slc13 ~]# nmap -p 53 172.17.0.153
Starting Nmap 7.70 ( https://nmap.org ) at 2024-08-26 09:16 UTC
Nmap scan report for SHRDC06.gasco.com (172.17.0.153)
Host is up (0.00013s latency).
```

```
PORT STATE SERVICE
53/tcp open domain

Nmap done: 1 IP address (1 host up) scanned in 0.49 seconds
[root@NWNA-SHR-slc13 ~]# nmap -p 88 gasco.com
Starting Nmap 7.70 ( https://nmap.org ) at 2024-08-26 09:16 UTC
Nmap scan report for gasco.com (10.127.38.24)
Host is up (0.00027s latency).
Other addresses for gasco.com (not scanned): 10.127.38.23 10.102.1.50 10.50.38.24 10.30.138.20 10.50.38.22 10.50.38.27 10.50.38.23 172.30.40.52 10.121.20.50 172.30.40.125 172.17.0.153 172.30.40.51 10.50.38.20 10.30.138.21
rDNS record for 10.127.38.24: SHRDC05.gasco.com
PORT STATE SERVICE
88/tcp open kerberos-sec
Nmap done: 1 IP address (1 host up) scanned in 0.47 seconds
Also got output for Powershell diagnostic:
Here is the credential screenshot:
and output of latest PS test script:

Questions:
Why credential test is getting completed in UI and failing when we do it via script?
Why we can see missed polls in data collection? Probably network issue?
How I can further investigate the issue?
```

### 2024-06-27 EM-65607: Powershell - Kerberos Timeout errors every 4 hours

ts_guide_to_jira_link

Link: EM-65607: Powershell - Kerberos Timeout errors every 4 hours CLOSED

From the description:

```
Observed Behavior:
Customers collectors are failing with the below error, every 4. This is impacting Windows collections.
2024-06-14T11:55:52+00:00 localhost powershell_collector[1601]: "2024-06-14 11:55:52,660","ERROR","ForkProcess-185","30973","ThreadPoolExecutor-1_20","139918600816384","powershell_pypsrp_transport","457","powershell_pypsrp_transport.communicate:
Failed to run command on did: 508194 - Maximum retries reached when trying to use kerberos to connect to server. Error: SpnegoError (4294967295): Major (851968): Unspecified GSS failure.  Minor code may provide more information, Minor (2529638944):
 Ticket expired, Context: Processing security token"

Expected Behavior:
Kerberos will assign new tickets as necessary and allow collections

Summary of troubleshooting steps taken so far and the current status:
Started on or around 21May, this doesn't correlate to anything they're aware of.
Concurrent PS and pypsrp are both enabled

Got customer on a WebEx and were able to get PS debug logging during a period of downtime, attached. We can see in the klist, image attached, the tickets for some reason have a 4 hour lifetime despite the default krb5.conf value of 24 hours.

Customer changed the krb5.conf file on one collector to have a ticket lifetime of 4 hours, which seemed to help, but I guess it wound up not actually working. Per slack thread with
```

```
Bryan Davis
https://sciencelogic.slack.com/archives/C0RGPQTED/p1719255049052989 it was suggested to increase the max kinit timeout value in the environment, as well as clean up some of the KDCs for this realm. Customer did both, and it has had no effect.

They have been unable to get any answers regarding if the problem exists on their side. Escalating to get guidance on figuring out where the problem is and how to fix it.

I will be OOO next week so will prep to transfer the support case.
```

### 2024-06-26 EM-65572: [LTIMindtree] Data collection issues for Windows servers from the data collectors through powershell

ts_guide_to_jira_link

Link: 💔 EM-65572: [LTIMindtree] Data collection issues for Windows servers from the data collectors through powershell CLOSED

From the description:

```
Observed Behavior:  the concurrent power-shell collection having issue: PowerShell Communication Error: [Credential ID: 645] ('Connection aborted.', ConnectionResetError(104, 'Connection reset by peer'))

Expected Behavior:  the collection should be working as expected.

Summary of troubleshooting steps taken so far and the current status:
Collected the logs and credential test status.
We can see it is failing within the container
The issue is not persistent if they are using legacy PS.
Tried re-configuring the device with winrm script as well.
This seems to be a similar issue as  https://sciencelogic.atlassian.net/browse/EM-65454#icft=EM-65454
```

### 2024-06-14 EM-65454: Concurrent PowerShell Collections Failing

ts_guide_to_jira_link

Link: EM-65454: Concurrent PowerShell Collections Failing CLOSED

From the description:

```
Observed Behavior: Get the following error on 5 devices (focused on one device): Error: ('Connection aborted.', ConnectionResetError(104, 'Connection reset by peer'))
Expected Behavior: No error should be seen and collections should occur

Summary of troubleshooting steps taken so far and the current status:
Behavior:

Legacy Collections work
Concurrent Powershell Collections fail
I have ran the powershell_test script inside and outside the container. It fails within the container
Issue appears not to be permissions as legacy collections work on this particular device.
From what I can tell, the collections stopped around the time of an upgrade back in January. Although difficult to know for sure
```

```
The attached archive has:powershell_diagnostic resultspowershell_test results from inside and outside the containerwinrm config from the end device
I have seen a Jira for the same error pertaining to a docker image however, the docker image in this case is relatively new. https://sciencelogic.atlassian.net/browse/EM-54212
I have also reviewed the system logs from the end device and can see the following entries:
A fatal alert was generated and sent to the remote endpoint. This may result in termination of the connection. The TLS protocol defined fatal alert code is 40.
Target name: %3
The TLS alert registry can be found at http://www.iana.org/assignments/tls-parameters/tls-parameters.xhtml#tls-parameters-6
and
An TLS 1.2 connection request was received from a remote client application, but none of the cipher suites supported by the client application are supported by the server. The TLS connection request has failed.
These are repeated 10 times each every 5 minutes
Attaching all collected data to the ticket
Also attaching screenshots of the TLS info from the end device
Discussed this with Engineering in Slack who requested the escalation - https://sciencelogic.slack.com/archives/C04L444TLG2/p1717663718458099
```

### 2024-06-12 EM-65410: powershell communication error

ts_guide_to_jira_link

Link: EM-65410: powershell communication error CLOSED

From the description:

```
Observed Behavior: Customer seeing Powershell communication errors for a specific workgroup.
Summary of troubleshooting steps taken so far and the current status:
Customer reported Powershell communication errors as seen in the screenshot attached. Credential test from UI shows successful and apps collecting data fine, but the performance graphs show no data. Please assist further diagnosing the issue.
```

### 2024-06-12 EM-65409: Collection stopped due to stale PowerShell service

ts_guide_to_jira_link

Link: EM-65409: Collection stopped due to stale PowerShell service CLOSED

From the description:

```
Observed Behavior:  PowerShell request are not processing the data with out restarting the PS service.
Expected Behavior: PowerShell request should populate the data without any issue.

Summary of troubleshooting steps taken so far and the current status:
The issue seems to be with the PowerShell services where the collection is neither completing nor erroring out.
Due to this there are no performance data for the devices.
Once we restart the PS services. the collection happening as expected.
this issue is not occurring on a regular basis. Hence client is unable to know when the issue is occurring. Hence they have created a RBA to restart the PS service every day for preventive measures
collected the diagnostic logs and debug logs.
Slack Ref: https://sciencelogic.slack.com/archives/C04L444TLG2/p1718006875360879
```

```
Ref Jira: https://sciencelogic.atlassian.net/browse/EM-55103#icft=EM-55103
```

### 2024-05-31 EM-65229: File system drives of windows are not showing

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Link: 💔EM-66624: Windows filesystems not discovered when SNMP v3 cred pwd is too short

CLOSED

From the description:

```
Summary:
Customer is not able to see the File system drive of Windows devices in the performance tab.
SL1 - 11.2.2
Microsoft: Windows Server v114

DID: 18736
Slack:- https://sciencelogic.slack.com/archives/CA16UME4T/p1715686648333649

We have run the DAs and could see they are able to collect the data.
Microsoft: Windows Server Disk Performance
Microsoft: Windows Server Disk Configuration
the only error that we see in the debug logs are

Cache miss, performing parent request, cache key = 0413_app_1790_did_18736_req_id_2200, parent_sync_key = 0413_app_1790_did_18736
powershell_application._execute_request: PowerShell error exception caught. Detail: Exception calling "Add" with "2" argument(s): "Item has already been added. Key in dictionary: 'broadcom netxtreme e-series advanced
dual-port 10gbase-t ethernet pcie adapter'  Key being added: 'broadcom netxtreme e-series advanced dual-port 10gbase-t ethernet pcie adapter'"At line:10 char:20Exception calling "Add" with "2" argument(s):
"Item has already been added. Key in dictionary: 'broadcom netxtreme e-series advanced dual-port 10gbase-t ethernet pcie adapter'  Key being added: 'broadcom netxtreme e-series advanced dual-port 10gbase-t ethernet pcie
adapter'"At line:11 char:23Exception calling "Add" with "2" argument(s): "Item has already been added. Key in dictionary: 'broadcom netxtreme e-series advanced dual-port 10gbase-t ethernet pcie adapter'  Key being
added: 'broadcom netxtreme e-series advanced dual-port 10gbase-t ethernet pcie adapter'"At line:12 char:20

Disk Performance is being collected fine, and they are not able to see the File Systems drop down on the performance page.

We could see SIGTERM and Process terminated on the collector monitoring the device 18736

"2024-05-27 06:27:02,588","ERROR","10140","1","proc_driver","867","Data Collection: Dynamic App process did not complete in its allowed timeslot, SIGTERM'ing. Run time: 27m (max 26m)"

"2024-05-27 06:28:01,178","ERROR","10140","1","proc_driver","867","Data Collection: Dynamic App process did not complete in its allowed timeslot, SIGTERM'ing. Run time: 28m (max 26m)"

"2024-05-27 06:28:01,512","INFO","67814","11","shared","280","Data Collection: Dynamic App task list at termination: [((597, 1236), 0), ((15965, 1236), 0), (15662, 1233), (18752, 1233), ((588, 1236), 0),
 ((18726, 1236), 0), ((18728, 1237), 0), ((18728, 1236), 0), (18785, 1233), ((5101, 1237), 0), ((1899, 1236), 0), ((18740, 1236), 0), ((14682, 1237), 0), ((5852, 1237), 0), ((587,
```

```
1237), 0), ((597, 1237), 0),
 (18764, 1233), ((15903, 1236), 0), ((14681, 1236), 0), ((15965, 1237), 0), ((18725, 1236), 0), ((14678, 1237), 0), (394, 1233), ((18736, 1236), 0), ((392, 1236), 0), (1110, 1233), (18817, 1233), ((1899, 1237), 0),
 ((14682, 1236), 0), ((14680, 1237), 0), ((5852, 1236), 0), ((15966, 1236), 0), ((18737, 1236), 0), ((14678, 1236), 0), ((14680, 1236), 0), (5858, 1233)]"

Not seeing too much details being logged in PowerShell logs

2024-05-30T11:10:52+00:00 localhost powershell_collector[1533]: "2024-05-30 11:10:52,763","INFO","MainProcess","8","MainThread","139898700171072","_main_","44","Starting PowerShell service loop..."
2024-05-30T11:10:52+00:00 localhost powershell_collector[1533]: "2024-05-30 11:10:52,764","INFO","MainProcess","8","MainThread","139898700171072","powershell_collector","2393","powershell_event_loop: Creating
tasks for receiving jobs, collection, and sending results ..."

2024-05-30T11:10:52.777254+00:00 bialsrv-slapp1 powershell_collector: powershell_collector-powershell_collector-1  | "2024-05-30 11:10:52,765","INFO","MainProcess","8","MainThread","139898700171072","powershell_collector",
"1451","collect_async: Getting all job bundles available (or waiting for one to be added)..."
```

The slack discussion is here: [https://app.slack.com/client/E03R7NUFC4X/CA16UME4T](https://app.slack.com/client/E03R7NUFC4X/CA16UME4T)

From that discussion @Bryan Davis noted:

```
The fix for the interface issue is in this ticket: https://sciencelogic.atlassian.net/browse/SOL-24724?focusedCommentId=358973.
Comment by Bryan Davis on a Support Defect in Closed
@Christian Lutz
Great.
Please have the customer follow these directions to update the request for the Interface Performance metrics.
1. In SL1, go to System->Manage->Dynamic Applications.
2. Search for “Windows Server Performance Cache”.
3. Wrench the Dynamic Application “Microsoft: Windows Server Performance Cache”.
4. Click on the PowerShell tab.
5. Wrench the PowerShell request “Windows Server Interface Performance”.
6. Blank out the current request in the text box and swap with the contents of the attached file “IntPerfRequest-Update.ps1”. When pasting in the new contents ensure there are no blank lines at the end. A blank line will cause the PowerShell command to fail as the system automatically appends commands to the end.
7. Click Save.
8. Close the window.
9. Wait approximately 5 minutes for a config push to occur. Attempt to run the Dynamic Application in debug mode and return the results. The correct data should appear.
Once confirmed this solves the problem I will convert this to a support defect.
#salesforce

```

### 2024-05-28: EM-65167: Powershell error

ts_guide_to_jira_link

Link: EM-65167: Powershell error CLOSED

From the description:

```
Reason for Escalation: Workaround required for high priority issue affecting customer
Which troubleshooting doc(s) were reviewed:
https://sciencelogic.atlassian.net/wiki/spaces/ENG/pages/12400825061/Powershell+Collection+Troubleshooting
https://sciencelogic.atlassian.net/wiki/spaces/ENG/pages/27500478877/PowerShell+Troubleshooting
https://sciencelogic.atlassian.net/wiki/spaces/ENG/pages/27425505298/Concurrent+PowerShell+Troubleshooting

Observed Behavior:PowerShell Communication Error: [Credential ID: 59] Forward and reverse name resolution is not properly configured for server, or the domain controller specified for OFFICE.CORP.INDOSAT.COM is not reachable
Expected Behavior:Powershell error should not appear.

Summary of troubleshooting steps taken so far and the current status:
1)Verified forward and reverse lookup is working for device and DNS server and resolving properly
2)verified /etc/krb5.conf and found no issue (attaching in the case)
3)kinit and klist is not throwing any error
4)Customer is using concurrent powershell and powershell_tests.py is throwing error"ERROR: authGSSClientStep() failed: (('Unspecified GSS failure.  Minor code may provide more information', 851968),
('Server not found in Kerberos database', -1765328377))"
5)attaching all the logs and screenshot
```

### 2024-05-21 EM-65087: KDC has no support for Encryption type - OL8 collectors

ts_guide_to_jira_link

Link: EM-65087: KDC has no support for Encryption type - OL8 collectors CLOSED

From the description:

```
Observed Behavior: PowerShell Communication Error: [Credential ID: 344] An error occurred updating the credential cache - kinit: KDC has no support for encryption type while getting initial credentials [return code = 1]
Expected Behavior: No errors from powershell collection

Summary of troubleshooting steps taken so far and the current status:
Andrew Doran Suggested Escalation
```

### 2024-05-20 EM-65082: Windows Servers of PURE-PURE FISHING is not showing performance data

ts_guide_to_jira_link

Link: EM-65082: Windows Servers of PURE-PURE FISHING is not showing performance data - credential empty CLOSED

From the description:

```
The Case 00436320 has been updated. To view the Case, please click here.
Note: You will need a Salesforce Account to click the above link.
Reason for Escalation: Engineering assistance needed to diagnose issue in order to find workaround
Which troubleshooting doc(s) were reviewed: https://sciencelogic.atlassian.net/wiki/spaces/KnowledgeB/pages/28628877313/Enhanced+Troubleshooting+Guide+-+PowerShell?search_id=b4f57b18-f77f-40bf-b5ca-4e16349733e0
Customer Timezone: APAC
Observed Behavior: Windows Servers of PURE-PURE FISHING is not showing performance data.
Expected Behavior: Windows Servers of PURE-PURE FISHING should show performance data.
Summary of troubleshooting steps taken so far and the current status:
Insight Shared - 02 - Hosted Windows Servers of PURE-PURE FISHING is not showing performance data
SL1 12.1.1 [build 605] OL 8
Microsoft: Windows Server v115
Concurrent PowerShell Collection Enabled

Found out that most (if not all) of PURE windows servers don't have data collection since April 28, 2024.

Debug log showing below:
2024-05-14 00:00:26,584","ERROR","MainProcess","6","MainThread","139621521731840","powershell_requests","258","Exception caught processing job for did 148284 and app Microsoft: Windows Server CPU Performance"
2024-05-14T00:00:26+00:00 localhost powershell_collector[1501]: Traceback (most recent call last):
2024-05-14T00:00:26+00:00 localhost powershell_collector[1501]: File "/usr/local/lib/python3.8/site-packages/powershell_collector/powershell_requests.py", line 240, in get_job_details
2024-05-14T00:00:26+00:00 localhost powershell_collector[1501]: cred_details = get_cred_details(
2024-05-14T00:00:26+00:00 localhost powershell_collector[1501]: File "/usr/local/lib/python3.8/site-packages/powershell_collector/powershell_requests.py", line 297, in get_cred_details
2024-05-14T00:00:26+00:00 localhost powershell_collector[1501]: cred_detail["cred_pwd"] = password_decode(found_cred.get("cred_pwd"))
2024-05-14T00:00:26+00:00 localhost powershell_collector[1501]: File "/usr/local/lib/python3.8/site-packages/powershell_collector/utilities.py", line 286, in password_decode
2024-05-14T00:00:26+00:00 localhost powershell_collector[1501]: raise EmptyPasswordError("No password was found for decoding")
2024-05-14T00:00:26+00:00 localhost powershell_collector[1501]: powershell_collector.exceptions.EmptyPasswordError: No password was found for decoding
2024-05-14T00:00:26.585309+00:00 pure-cld-slc01 powershell_collector: powershell_collector-powershell_collector-1 | "2024-05-14 00:00:26,584","ERROR","MainProcess","6","MainThread","139621521731840","powershell_requests","258","Exception caught processing job for did 148284 and app Microsoft: Windows Server CPU Performance"
2024-05-14T00:00:26.585570+00:00 pure-cld-slc01 powershell_collector: powershell_collector-powershell_collector-1 | Traceback (most recent call last):
2024-05-14T00:00:26.585785+00:00 pure-cld-slc01 powershell_collector: powershell_collector-powershell_collector-1 | File "/usr/local/lib/python3.8/site-packages/powershell_collector/powershell_requests.py", line 240, in get_job_details
2024-05-14T00:00:26.585983+00:00 pure-cld-slc01 powershell_collector: powershell_collector-powershell_collector-1 | cred_details = get_cred_details(
2024-05-14T00:00:26.586193+00:00 pure-cld-slc01 powershell_collector: powershell_collector-powershell_collector-1 | File "/usr/local/lib/python3.8/site-
```

```
packages/powershell_collector/powershell_requests.py", line 297, in get_cred_details
2024-05-14T00:00:26.586388+00:00 pure-cld-slc01 powershell_collector: powershell_collector-powershell_collector-1 | cred_detail["cred_pwd"] = password_decode(found_cred.get("cred_pwd"))
2024-05-14T00:00:26.586581+00:00 pure-cld-slc01 powershell_collector: powershell_collector-powershell_collector-1 | File "/usr/local/lib/python3.8/site-packages/powershell_collector/utilities.py", line 286, in password_decode
2024-05-14T00:00:26.586787+00:00 pure-cld-slc01 powershell_collector: powershell_collector-powershell_collector-1 | raise EmptyPasswordError("No password was found for decoding")
2024-05-14T00:00:26.586981+00:00 pure-cld-slc01 powershell_collector: powershell_collector-powershell_collector-1 | powershell_collector.exceptions.EmptyPasswordError: No password was found for decoding

Manually run the DA Microsoft: Windows Server CPU performance for more than 5 minutes and it seems to keep looping and not going any way...

Have re-saved the 5 credentials but issue still persist.
Debug and diagnostic logs attached including new DE-cred-powershell-2024-05-20.tar.gz after credentials re-saved.

Discussion Slack Channel #team-collectables
· https://sciencelogic.slack.com/archives/CA16UME4T/p1715781216180759
```

### 2024-05-14 EM-64965: PowerShell Error for devices in ScienceLogic Events

ts_guide_to_jira_link

Link: EM-64965: PowerShell Error for devices in ScienceLogic Events CLOSED

From the description:

```
Observed Behavior:
Expected Behavior:
Summary of troubleshooting steps taken so far and the current status:
Frequent notice alert - PowerShell Communication Error: App [850] Windows device [10.36.38.230]: authGSSClientStep() failed: (('Unspecified GSS failure. Minor code may provide more information', 851968), ('Ticket expired', -1765328352))
DA - Windows Server configuration Cache.
Windows PP version - 116
SL1 version 12.1.1
Concurrent is enabled DA is collecting fine without any issues, but the alert is frequently coming for some of the devices.
https://sciencelogic.slack.com/archives/CA16UME4T/p1715103377052699
```

### 2024-05-07 EM-64874: Concurrent power-shell issue

ts_guide_to_jira_link

Link: EM-64874: Concurrent power-shell issue CLOSED

From the description:

```
Observed Behavior: No performance data collection is happening when we enable concurrent PowerShell.
Expected Behavior: Collection should work fine on concurrent PowerShell.
Summary of troubleshooting steps taken so far and the current status:
```

```
No performance data collection is happening when we enable concurrent PowerShell.
Windows PP version 116
SL1 version 12.1.1
Credential test is getting passed. - attached the screenshot
Attached the powershell_diagnostics.py output.
```

### 2024-05-03 EM-64811: PowerShell communication errors on windows devices

ts_guide_to_jira_link

Link: EM-64811: PowerShell communication errors on windows devices CLOSED

From the description

```
Summary of troubleshooting steps taken so far and the current status:
Concurrent PowerShell
Windows PP version: v116
When we testing the credential test , we're getting the following error - . PowerShell Communication Error: App [727] Windows device [10.10.100.139]: authGSSClientStep() failed: (('Unspecified GSS failure. Minor code
may provide more information', 851968), ('KDC reply did not match expectations', -1765328237))

The task was successful on the other collector in the same collector group. I've verified the krb5.conf file and the user permissions, and everything appears to be in order. We even created a new user and aligned the role,
but unfortunately, we're still facing issues. I've gathered all the relevant information. Could you please review it and let me know what might be causing the problem?
```

### 2024-04-15 EM-64394: Proxy WinRM configuration is not passing the credential test

ts_guide_to_jira_link

Link: EM-64394: Proxy WinRM configuration is not passing the credential test CLOSED

From the description:

```
Observed Behavior: False alert - PowerShell Communication Error: Forward and reverse name resolution is not properly configured for server, or the domain controller
specified for IPTMGMT.LOCAL is not reachable.
Expected Behavior: False alerts should not trigger on the server.
```

### 2024-04-10 EM-64261: Windows Server Failing Credential Test with HTTP Code 400

ts_guide_to_jira_link

Link: EM-64261: Windows Server Failing Credential Test with HTTP Code 400 CLOSED

From the description:

```
Observed Behavior: Windows server showing "Bad HTTP response returned from server. Code 400(WinRMTransportError: Bad HTTP response from server . Code 400.)" error during credential test.
Expected Behavior: Credential test should be passing so the Windows server can be discovered properly.
```

```
Summary of troubleshooting steps taken so far and the current status:
Customer reports that when running a credential test on a Windows server, they are getting the error:
Bad HTTP response returned from server. Code 400(WinRMTransportError: Bad HTTP response from server . Code 400.)

This seems to match the previous Jiras:
https://sciencelogic.atlassian.net/browse/EM-58309
https://sciencelogic.atlassian.net/browse/EM-58320
https://sciencelogic.atlassian.net/browse/EM-61211

However, all of the previous fixes for the Jiras above are all for SL1 11.3.x, whereas the customer is still on SL1 11.2.1. I am forwarding this to engineering to confirm if this is in fact the
same issue and, if it is, can a drop file be created for this version of SL1. If it is not, can a fix be provided for the customer.
```

### 2024-03-12: EM-63731: PowerShell request error: App [917] Request [929]: Get-ChildItem : Requested registry access is not allowed. At line:1 char:56

ts_guide_to_jira_link

Link: EM-63731: PowerShell request error: App [917] Request [929]: Get-ChildItem : Requested registry access is not allowed. At line:1 char:56 CLOSED

From the description:

```
Observed Behavior: PowerShell request error: App [917] Request [929]: Get-ChildItem : Requested registry access is not allowed. At line:1 char:56
Expected Behavior: As per the given permission, the poweshell collection should work fine.
Summary of troubleshooting steps taken so far and the current status:
SL1 is trying to execute following command from DA: Microsoft: Windows Server IC Process Service Cache and Snippet: Windows Server Service Details - ICDA
$triggerInfo = Get-ChildItem - Path:Registry::HKLM\SYSTEM\CurrentControlSet\Services*\triggerinfo | Select @{Expression={$_.Name.Split("\")[4]};Label="Name"}
Error we're getting: PowerShell request error: App [917] Request [929]: Get-ChildItem : Requested registry access is not allowed. At line:1 char:56
We followed the document steps - Configuring the User Account for the ScienceLogic Platform
Still we're not able to collect the services related data.
We also aligned the SL1 user to read the read the following key and all its descendants: HKLM\SYSTEM\CurrentControlSet\Services. (Screenshots attached)
Output of command in locally on windows server where lot of initial lines are empty (Screenshots attached).
We also verified the user domain is part of local users group.
```

### 2024-02-16 EM-63374: PowerShell Communication errors across multiple devices. Missing reason in the error message

ts_guide_to_jira_link

Link: EM-63374: PowerShell Communication errors across multiple devices. Missing reason in the error message CLOSED

From the description:

```
Observed Behavior: Customer has intermittent PowerShell collection issues and the error message does not contain the reason why collections are failing.

Expected Behavior: PowerShell collections should be stable and not fail. Whenever PowerShell collection fails, an error message indicating the problem can help find out the root cause.
```

### 2024-02-05 EM-63205: [hosted-shared7] powershell error updating credential cache

ts_guide_to_jira_link

Link: EM-63205: [Hosted-Shared7] Powershell error updating credential cache CLOSED

From the description

```
OK... I'm out this week, but what this is showing is an issue with libem7 which is a library owned by syseng.
The test script is getting the same error as the container when decrypting the password.When the pass word is
provided as an input to the test script, it works as libem7 is not used. When the credid is used the encrypted
password is read and libem7 used to decrypt and that's where the "empty password' message comes from.You'll
need to escalate and syseng will need to either have the escalations or help out collectables with it
(its not a 50c issue)
```

From this comment EM-63205: [Hosted-Shared7] Powershell error updating credential cache CLOSED

```
 now believe the Credential Gateway Service socket may not be mounted into the container, which would explain the observations as I understand them.  I’ve started a thread in Proc’n’Roll’s team
 channel, both to see if I’ve read things correctly and to find out if this is expected, as one of the phases of roll-out for the CyberArk connector.  Once @Randy Murdock and crew confirm one
 way or the other, I think we’ll have our answer, which I’d imagine would be one of:
    This is known behavior, which is planned to be fixed by de-containerization anyway, or
    This is a defect, which will be fixed by de-containerization, but which could also warrant a compose file adjustment as an interim fix
If it’s the latter, I’d suspect the fix would be applied by 50CentOS and/or Proc’n’Roll, though I’d be happy to help as needed.
```

and from this comment

```
k, it is case (1), with CyberArk connectivity having been designed for and tested with standard discovery and collections only, i.e. excluded from containerized collections in the first phase of CyberArk integration.
It is technically possible to add a socket mount for the CGS to containers' configurations, but we do not have a production configuration for this yet.  I would consider that a feature request, t
hough I’d also imagine CSM could negotiate it up to a defect if there’s widespread need.
@Eric DeRose this is expected behavior, but probably deserves a PM conversation.
@Ketan Gharate I imagine you’ll want to touch base with the customer’s CSM before forming a response, based on this conclusion, since it is a somewhat adverse result.  Please let me know if I can
help explain the findings further.
```

and from another comment:

```
Disabling Concurrent PowerShell is the solution, and this solution is already available in the version of SL1 the customer is using.
When Concurrent PowerShell is disabled, SL1 will use standard collection processes for PowerShell collections, and standard collection processes are integrated with CyberArk credential management.
Containerized collection services will be de-containerized in a future release, after which their integration with CyberArk will be reevaluated.  There is no set timeline for this.
```

### 2023-09-30 EM-61211: We are unable to discover windows servers

ts_guide_to_jira_link

Link: EM-61211: We are unable to discover windows servers CLOSED

From the description:

```
Summary:
Customer is running a credential test and getting the error
Bad HTTP response returned
from server. Code 400 (WinRMTransportError: Bad HTTP response returned from server. Code 400)- Not using encrypted [want to use unencrypted way]

    Group privileges are fine

    We have also run winrm_configuration_wizard to ensure there is no issue with permissions

    It is AD account

    Not seeing too much of details in the Event Viewer logs
    Windows Remote Management  >  Operational

    We have also restarted winrm service

    We Tried with the below options as well, but the result was the same
    WSMAN://%D
    HOST://%D
    - AllowUnencrypted is set to true, and we have also restarted the winrm service
```

### 2023-04-30 EM-58320: PowerShell Communication Error: [Credential ID: xx] 'powershell_pywinrm_transport' object has no attribute 'realm' - Container Updates

ts_guide_to_jira_link

Link: EM-58320: PowerShell Communication Error: [Credential ID: xx] 'powershell_pywinrm_transport' object has no attribute 'realm' - Container Updates CLOSED

From the description:

```
Summary:
Loads of devices seeing these errors after upgrading to 11.3.1, 97 devices are reporting these errors:
PowerShell Communication Error: Credential ID: 233 powershell_pywinrm_transport object has no attribute realm (message repeats 31 times)
```

```
PowerShell Communication Error: App [1303] Windows device 10.29.70.121: Bad HTTP response returned from server. Code 400 (WinRMTransportError: Bad HTTP response returned from server. Code 400)
Andrew Doran knows about the issue:
The “realm” error is a bug - I did ask that a Jira ticket be created for that. It’s a bug in the error handling - the python script should confirm the root cause error if run against that device.So there is
an underlying issue (looks like in this case the 400 error) but when we are trying to turn that into an event, we fail because we are using an invalid property. We need to fix that error handling so the
event is properly handled - but it’s not the underlying issue
Workaround:
N/A
Steps to reproduce:
upgrade to 11.3.1
```

### 2022-11-22 EM-54839: Exception caught while communicating

ts_guide_to_jira_link

link: SOL-28091: Exception caught while communicating CLOSED

This error:

```
Microsoft PowerShell: [DID: 53687 User: svc-sciencelogic] Exception caught while communicating with http://SERVERNAME.DOMAIN:5985wsman (retrying connection)
```

was being seen for 36 of the 143 windows servers being monitored.

This was resolved by

```
Adjusting the timeout in the credential to 30000ms fixed this
```

This shows some of the details of the error messages seen:

```
2022-11-11T00:00:21+00:00 localhost powershell_collector[1233]: "2022-11-11 00:00:21,475","INFO","ForkProcess-6503","28290","ThreadPoolExecutor-0_42","140600225515264","kerberos_ticket","53","KerberosTicketMgr.update_krb5_info: Opening subprocess as user root to run "kinit -c /tmp/krb5cc_CORP.INTERNALROOT.COM_svc-sciencelogic svc-sciencelogic@CORP.INTERNALROOT.COM""
2022-11-11T00:00:21+00:00 localhost powershell_collector[1233]: "2022-11-11 00:00:21,495","INFO","ForkProcess-6503","28290","ThreadPoolExecutor-0_42","140600225515264","kerberos_ticket","53","KerberosTicketMgr.update_krb5_info: Calling communicate() to store ticket."
2022-11-11T00:00:21+00:00 localhost powershell_collector[1233]: "2022-11-11 00:00:21,515","INFO","ForkProcess-6503","28290","ThreadPoolExecutor-0_42","140600225515264","powershell_transports","716","_call_pywinrm [connect]: Opening remote shell on http://UKTEDBASQL03D.veecommerce.net:5985/wsman for user svc-sciencelogic\CORP.INTERNALROOT.COM"
2022-11-11T00:00:21+00:00 localhost powershell_collector[1233]: "2022-11-11 00:00:21,523","ERROR","ForkProcess-6503","28290","ThreadPoolExecutor-0_42","140600225515264","kerberos_","257","generate_request_header(): authGSSClientStep() failed:"
2022-11-11T00:00:21+00:00 localhost powershell_collector[1233]: Traceback (most recent call last):
2022-11-11T00:00:21+00:00 localhost powershell_collector[1233]:   File "/usr/local/lib/python3.8/site-packages/winrm/vendor/requests_kerberos/kerberos_.py", line
```

```
245, in generate_request_header
2022-11-11T00:00:21+00:00 localhost powershell_collector[1233]:     result =
kerberos.authGSSClientStep(self.context[host],
2022-11-11T00:00:21+00:00 localhost powershell_collector[1233]: kerberos.GSSError:
(('Unspecified GSS failure.  Minor code may provide more information', 851968), ('Server not
found in Kerberos database', -1765328377))
2022-11-11T00:00:21+00:00 localhost powershell_collector[1233]: "2022-11-11
00:00:21,523","ERROR","ForkProcess-6503","28290","ThreadPoolExecutor-
0_42","140600225515264","kerberos_","259","(('Unspecified GSS failure.  Minor code may
provide more information', 851968), ('Server not found in Kerberos database', -1765328377))"
2022-11-11T00:00:21+00:00 localhost powershell_collector[1233]: Traceback (most recent call
last):
2022-11-11T00:00:21+00:00 localhost powershell_collector[1233]:   File
"/usr/local/lib/python3.8/site-packages/winrm/vendor/requests_kerberos/kerberos_.py", line
245, in generate_request_header
2022-11-11T00:00:21+00:00 localhost powershell_collector[1233]:     result =
kerberos.authGSSClientStep(self.context[host],
2022-11-11T00:00:21+00:00 localhost powershell_collector[1233]: kerberos.GSSError:
(('Unspecified GSS failure.  Minor code may provide more information', 851968), ('Server not
found in Kerberos database', -1765328377))
2022-11-11T00:00:21+00:00 localhost powershell_collector[1233]: "2022-11-11
00:00:21,523","ERROR","ForkProcess-6503","28290","ThreadPoolExecutor-
0_42","140600225515264","powershell_transports","868","_call_pywinrm [connect]: [DID: 64084
User: svc-sciencelogic] Exception caught while communicating with
http://UKTEDBASQL03D.veecommerce.net:5985/wsman (retrying connection) - detail:
authGSSClientStep() failed: (('Unspecified GSS failure.  Minor code may provide more
information', 851968), ('Server not found in Kerberos database', -1765328377)) "
2022-11-11T00:00:21+00:00 localhost powershell_collector[1233]: "2022-11-11
00:00:21,524","ERROR","ForkProcess-6503","28290","ThreadPoolExecutor-
0_42","140600225515264","powershell_transports","890","_call_pywinrm [connect]:  ([DID:
64084 User: svc-sciencelogic] Unexpected exception occurred:  authGSSClientStep() failed:
(('Unspecified GSS failure.  Minor code may provide more information', 851968), ('Server not
found in Kerberos database', -1765328377)) )"
2022-11-11T00:00:21+00:00 localhost powershell_collector[1233]: "2022-11-11
00:00:21,524","ERROR","ForkProcess-6503","28290","ThreadPoolExecutor-
0_42","140600225515264","powershell_app_collection","257","PowerShellCollection._connect:
Error caught establishing connection using IP 10.128.165.103: Windows device
[10.128.165.103]: Forward and reverse name resolution is not properly configured for server,
or the domain controller specified for CORP.INTERNALROOT.COM is not reachable."
```

### 2019-01-09 EM-34627: Windows server denying access for powershell credentials.

Link: SOL-17812: [HCL - DryIce] - Windows server denying access for powershell credentials.

CLOSED

ts_guide_to_jira_link

The customer lost interest before this was resolved.